Anomaly and Interestingness Detection in Timed Hierarchical Business Processes

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***Abstract*—Anomaly detection is assumed to be an efficient means of probing cyber-attacks upon business processes, and frequent pattern recognition (referred as interestingness) is complementary to anomaly detection for providing insights about processes. Current studies on anomalies and interestingness focus mostly on measurement design and discovery methods. However, parameterized evaluation from a time dimension has not been well-explored to satisfy personalized demands of edge users. To fill this gap, this article proposes to detect temporal anomalies and interestingness leveraging time-constrained and granularity-aware signatures. Specifically, a *Timed Hierarchical Business Process Model* (*TH-BPM*) is constructed by exploiting temporal constraints and granularities thoroughly, and composing activities from fine granularities to coarser ones. Temporal anomalies are estimated with prescribed signatures upon *TH-BPM* through parameterizing acceptance of deviant executions. Temporal interestingness, as the complement to temporal anomalies, is specified as the most probable execution time that is partitioned into user-defined granules and ranked by the probability leveraging the parameterized signatures. Consequently, various acceptance of anomaly deviations is allowed, and parameterized needs of interestingness detection can be satisfied. Experimental results upon real-life event logs and a real edge network demonstrate that our approach outperforms the state-of-the-art techniques in relevant performance metrics.**

***Index Terms*—Parameterized signature, temporal anomaly, temporal interestingness, timed hierarchical business processes.**

I. Introduction

CYBERSECURITY is defined as a set of processes, human behavior, and related systems that protect against potential cyber-attacks [1]. Techniques for identifying patterns that do not conform to expected behaviors (a.k.a., *anomalies*) [2], or that inferring frequent behaviors (referred as *interestingness*, which is the complement to anomalies) [3], have been proposed for protecting the cybersecurity in recent literature [4]. Detecting *anomalies* in business processes, which are composed of a set of partially ordered activities [5], is of significant importance to alert potential cyberthreats or fraudulent behaviors [6], [7]. Interestingness detection, which is mostly relevant to subjective perception [8] has been explored to manifest certain statements prescribed by users. This article refers to the most frequent behaviors of business processes as *interestingness* [9]. Traditional techniques have been proposed to detect anomalies and interestingness with various measurement design and discovery algorithms [10], [11], whereas an evaluation from a temporal aspect has not been explored extensively [12]. In fact, business activities are time-constrained, such that certain time constraints have to be complied with [13]. This characteristic leads to the detection of temporal anomalies and interestingness. In this article, we are motivated to achieve this detection with the modeling of time-constrained business activities.

Techniques have been developed to investigate temporal constraints for processes from intra- and interactivity aspects [14], [15], including time-aware Cloud resource allocation for business processes [16], service composition for timed applications in *Internet of Things* (*IoT*) [17], temporal network representation of event logs [18], [19], modeling of discrete event systems with temporal constraints [20], identification of temporal requirements for process modelling languages [21], verification of business process models as timed automata [22], and investigation of violation of temporal process constraints [23]. enerally, these efforts attempt to explore certain temporal constraints involved in business activities [24], to achieve versatile temporal analysis goals. However, they fall short of achieving a comprehensive investigation of temporal constraints as presented by the evaluation shown in Table II, and few efforts were made on that of collaborative activities. Besides, temporal granularity is not well-explored currently, although it reveals valuable information in business processes [25]. In fact, time stamps in event logs may have various levels of granularities ranging from seconds in precision to coarse granules. Based on partially specified temporal constraints, temporal modeling and analysis are limited. To fill this gap, this article attempts to explore temporal constraints and granularities for individual activities, and also collaborative ones and their connecting edges, to facilitate a thorough investigation of temporal anomalies and interestingness.

To address the detection challenge of anomalies and interestingness, efforts have been proposed using outlyingness- and signature-based detection methods. Outlyingness-based detection techniques rely on classification models trained by using historical data to discover anomalous behaviors [26], [27]. They are perceived as powerful due to higher potential to identify

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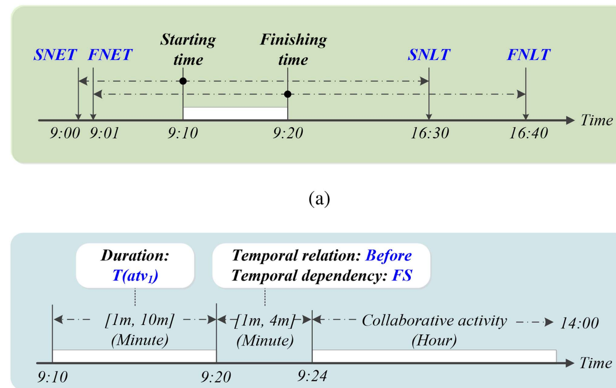
novel anomalies, but they constantly face higher false positive rate. Nevertheless, they require a training phase and consume a large amount of computing resources [28], [29], which make them difficult to be applied on edge devices, since edge devices are usually limited in computational and storage resources [30]. In contrast, signature-based techniques show advantages on a more effective detection leveraging pattern recognition techniques [31]. They are conducted by developing a reference model serving as a formal representation of normal behaviors. Anomalous behaviors, which deviate from the model, are recognized. Generally, these studies depend on preconfigured signatures (rules, or patterns) [32]. Due to the application complexity and personalized demands, anomalous (or interesting) executions may not be well defined or cannot be distinguished by a same signature, such as setting a threshold to deal with infrequent behaviors. Therefore, a high level of flexibility to satisfy parameterized requirements is of significance in real scenarios.

To address these challenges, a comprehensive modeling of temporal constraints and granularities is proposed, and temporal anomalies and interestingness by parameterizing time-constrained and granularity-aware signatures are detected. The contributions of this article are summarized as follows:

1) A *Timed Hierarchical Business Process Model* (*TH-BPM*) is proposed through a comprehensive investigation of temporal constraints and granularities. Specifically, i) starting and finishing time, and durations of intraactivities, and ii) temporal relations and dependencies of interactivities, are explored. Afterward, timed activities are composed to collaborative ones considering temporal granularities, and are modeled in a hierarchical manner.
2) We conduct a customized detection for temporal anomalies based on time-constrained and granularity-aware signatures, with respect to timed individual activities and collaborative ones, and their connecting edges upon *TH-BPM*. Temporal anomalies are detected by matching prescribed signatures, where different acceptance of deviant executions is enabled by parameterizing signatures.
3) In a similar manner, personalized requirements for temporal interestingness can be satisfied by signature-based parameterization. Temporal interestingness is formulated as the most probable execution time for timed activities and edges upon *TH-BPM*, where the execution time can be partitioned into user-defined granules and ranked by their probabilities leveraging prescribed signatures.

Extensive experiments are conducted upon public real-life event logs. Evaluation results demonstrate the outperformance of our technique in terms of *accuracy*, *efficiency*, and *generality*, in comparison with the state-of-the-art techniques. Specifically, our method improves the accuracy by 6.08% compared with baseline techniques, and reduces execution time to less than 50 ms in heterogenous scenarios.

The remainder of this article is organized as follows. Section II presents relevant concepts. Section III introduces our motivating example. Section IV defines the timed hierarchical business process model. Section V elaborates the detection of temporal anomalies and interestingness. Extensive experiments are conducted in Section VI. Section VII discusses related techniques. Finally, Section VIII concludes this article.



(a)

(b)

Fig. 1. Examples of absolute and relative temporal constraints.(a) Example 1: Absolute Temporal Constraints (in Definition 1). (b) Example 2: Relative Temporal Constraints (in Definition 2).

II. Temporal Constraint and Granularity

This section presents the notions of temporal constraint and granularity. Temporal constraints are usually specified with respect to business activities, and can be both absolute and relative [9]. We refer the readers to [13] more details.

Definition 1 (Absolute Temporal Constraints): Specify starting and finishing time of intraactivities as follows:

1) *Starting Time:* Starting time of an activity can be constrained as *S*tart *N*o *E*arlier *T*han (*SNET*), *S*tart *N*o *L*ater *T*han (*SNLT*), or *M*ust *S*tart *O*n (*MSO*) (i.e., *SNET* equals *SNLT*), a certain absolute time point.
2) *Finishing Time:* Finishing time can be constrained as *F*inish *N*o *E*arlier *T*han (*FNET*), *F*inish *N*o *L*ater *T*han (*FNLT*), or *M*ust *F*inish *O*n (*MFO*) (i.e., *FNET* equals *FNLT*), a certain absolute time point.

Example 1 (Absolute Temporal Constraints): A sample activity atv with its starting and finishing time is illustrated in Fig. 1(a), i.e., atv starts at 9:10, and finishes at 9:20. Its starting time is constrained by *SNET* 9:00 and *SNLT* 16:30, while its finishing time is constrained by [*FNET*, *FNLT*]: [9:01, 16:40].

Definition 2 (Relative Temporal Constraints): Specify requirements for activity duration, temporal relation, and temporal dependence of interactivities, as follows:

1) *Activity Duration:* An activity atv is usually registered with its starting time and finishing time, and is expressed with a time duration $T(atv)$. Formally, $T(atv)$ is constrained within a range [*MinT*, *MaxT*].
2) *Temporal Relation:* Temporal relations investigate constraints crossing the execution boundaries of activities. An activity may depend on the starting or finishing time of another to begin or terminate. Thereafter, an activity atv_j can be executed *after, met by, equals, overlapped by, contains, started by* or *finished by* another atv_i [18].
3) *Temporal Dependence:* Temporal dependencies quantify temporal relations with real values for *S*tart-to-*S*tart (*SS*), *S*tart-to-*F*inish (*SF*), *F*inish-to-*S*tart (*FS*) and *F*inish-to-*F*inish (*FF*). Specifically, (i) *SF*(atv_i, atv_j) is exploited when atv_j is executed *after, met by*, or *overlapped by* atv_i, (ii) *FS*(atv_i, atv_j) is calculated when atv_j is executed *after*

atv_i, (iii) *SS*(atv_i, atv_j) and *FF*(atv_i, atv_j) are studied when atv_i *overlaps* atv_j, and (iv) the total execution time (i.e., T(atv_j)) of atv_i and atv_j is explored when atv_j *equals, contains, started by* or *finished by* atv_i.

Example 2 (Relative Temporal Constraints): Fig. 1(b) illustrates an activity with its duration, i.e., *T*(atv_1) = 10 min, which is constrained within a range [*MinT*, *MaxT*] = [*1m, 10m*]. In this example, atv_1 is executed *before* a collaborative activity, and thus, *FS* is calculated, e.g., *FS* = [*1m, 4m*], which means that the collaborative activity is executed after atv_1 by 1 to 4 min,. Besides, *SF* is exploited (i.e., from 9:10 to 14:00, *4h50m*) to infer their total time consumption.

Furthermore, temporal granularity, as an integral feature of temporal data, is of significance for temporal modeling and business process analysis [25]. Formally, a set of granularities can be defined from a calendar as follows.

Definition 3 (Temporal Granularities): Temporal granularities are specified from the calendar *cad* = (*Gra*, *MappingFunc*), where

1) *Gra* is a set of granularities partitioned from time line, where *Gra* = $\{gra_1, gra_2,..., gra_k\}$ (k = 1, 2,..., g).
2) *MappingFunc* is a collection of mapping functions between two granularities gra_{k_i} and gra_{k_j} in *Gra*.

This article adopts the widely used Gregorian calendar to describe the granularities that commonly appear in event logs. Specifically, *cad.Gra* = {*millisecond*, *second*, *minute*, *hour*, *day*, *week*, *fortnight*, *month*, *year*} (g = 9), which is abbreviated as $\{ms, s, m, h, d, w, fn, mth, y\}$. Mapping functions can be specified with two arguments of granularities to indicate their mapping relations and values, which are presented as follows:

1) *Mapping relations between granularities* can be specified by *MappingFuncRel* $(gra_{k_i}, gra_{k_j}) = k_j - k_i \in \mathbf{N}$, where $gra_{k_i} \in$ *Gra* and $gra_{k_j} \in$ *Gra* are two arguments.
2) *Mapping values between granularities* can be specified by *MappingFuncVal* $(gra_{k_i}, gra_{k_j}) = val_{ij} \in \mathbf{R}$, where $gra_{k_i} \in$ *Gra* and $gra_{k_j} \in$ *Gra* are two arguments.

Example 3 (Mapping Functions): For *hour* and *day*, (i) *MappingFuncRel* (*hour*, *day*) = 1, and *MappingFuncRel* (*day*, *hour*) = −1, meaning that *day* is coarser than *hour*, and (ii) *MappingFuncVal* (*hour*, *day*) = 24, and *MappingFuncVal* (*day*, *hour*) = -24, indicating that 24 hours constitute one day.

Leveraging the mapping functions, we define (i) the time duration of a granularity gra_{k_i} by $G\,(gra_{k_i})$ = *MappingFuncVal* (gra_{k_i}, $gra_{k_{i+1}}$), where gra_{k_i} and $gra_{k_{i+1}} \in$ *Gra*, and (ii) the interval of gra_{k_i} by *Itv* (gra_{k_i}) = *MappingFuncVal* ($gra_{k_{i-1}}$, gra_{k_i}), where $gra_{k_{i-1}}$ and $gra_{k_i} \in$ *Gra*.

Example 4 (Duration and Interval of a Granularity): *G* (*hour*) = *MappingFuncVal* (*hour*, *day*) = 24 hours, and *Itv* (*hour*) = *MappingFuncVal* (*minute*, *hour*) = 60 min.

III. Motivating Example

This section introduces our motivating example from a Telco operator [33], where a sample business process model is generated from a real-life event log, and represented in terms of *BPMN* as shown in the bottom of Fig. 2. This process deals with complaint signals raised by a customer as follows:

Example 5 (Business Process): An expert requests a "*Get Service Trouble Ticket*" (denoted atv_1) once a signal is received due to quality decrease or an exception. Then, a "*Retrieve Data*" (denoted atv_2) or a "*Start Service Test Management*" (denoted atv_3) is launched. In the case of data retrieval, the expert can choose "*Perform an Automated Retrieval*" (denoted atv_4) or "*Perform a Scripted Retrieval*" (denoted atv_5). The service test management starts by atv_3, and is implemented by a "*Start Scripted Service Test*" (denoted atv_6) or a "*Start Automated Service Test*" (denoted atv_7). The process terminates with a "*Reply to Customer*" (denoted atv_8) and a "*Trouble Shoot the Process*" (denoted atv_9).

Identifying anomalous and interesting patterns from business processes, referred as the detection of temporal anomalies and interestingness, shows significance in scenarios like cyberattacks protection. Experts should be informed of anomalous behaviors (as exemplified as follows), to prevent potential cyberthreats, or possible quality decrease:

Example 6 (Temporal Anomaly): A temporal anomaly appears when the duration of an automated retrieval is much longer than expected, i.e., *T*(atv_4) $\gg atv_4$.*MaxT*.

In consideration of the case that atv_4 takes a certain time between [atv_4.*MinT*, atv_4.*MaxT*], how far *T*(atv_4) is away from anomalous behaviors may be concerned as interestingness.

Example 7 (Temporal Interestingness): Whether the duration *T*(atv_4) lies in a certain refined range with a high probability (e.g., 90%) is studied as temporal interestingness.

Besides exploiting the executions of individual activities, experts might be concerned with i) collaborative ones, for studying behaviors of multiple activities in an integrated manner, and ii) their connecting edges, as shown in the following example. A thorough investigation of temporal constraints is proposed to facilitate detecting temporal anomalies and interestingness.

Example 8 (Temporal Constraint): The activities atv_4 and atv_5 may be concerned collaboratively in the case of data retrieval. The interval between atv_1 and atv_2 may be studied.

Temporal granularity, as an important aspect of temporal information, can improve the effectiveness in detecting temporal anomalies and interestingness. Specifically, it distinguishes the executions of business activities by their granularities in *Gra*.

Example 9 (Temporal Granularity): As shown in Fig. 2, the granule of *T*(atv_4), denoted as *T*(atv_4).*gra* = *hour* $\in$ *Gra*, is coarser than that of the majority. As the result, experts may be concerned with atv_4, since it consumes much more time.

In a similar manner, if an anomaly appears with a deviation, a coarser granularity of the deviation can make such an anomaly be identified more easily.

Example 10 (Temporal Granularity for Temporal Anomaly): Given a deviation *dvt* = *T*(atv_4) − atv_4.*MaxT* where *T*(atv_4) > atv_4.*MaxT*, *dvt.gra* equaling to *hour* rather than *minute* can make such an anomaly more easily to be identified.

In real applications, personalized detection of temporal anomalies and interestingness is increasingly demanded.

Example 11 (Personalized Detection of Temporal Anomaly): A slightly longer execution time (e.g., *dvt* is 10 s) can be alternatively reported as an anomalous event according to user's customized acceptance of deviations.

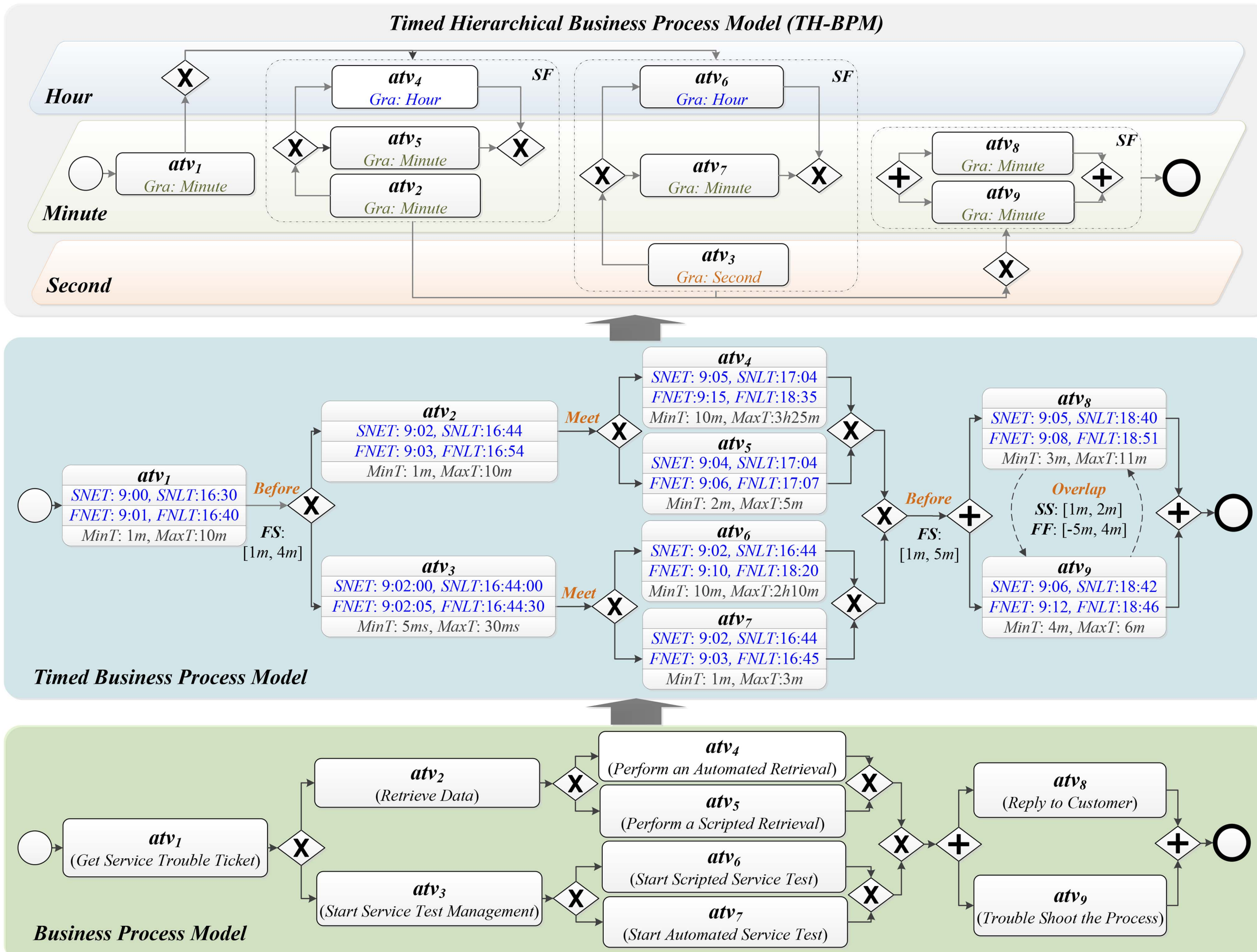


Fig. 2. Sample timed hierarchical business process model is presented as our motivating example, where the business process model is generated and represented in *BPMN* format as shown in the bottom part. Temporal constraints for intra- and interactivities are explored, where timed activities and collaborative ones are evaluated with starting time (constrained by *SNET*, *SNLT*, or *MSO*), finishing time (constrained by *FNET*, *FNLT*, or *MFO*) and durations in terms of [*MinT*, *MaxT*], as illustrated in the middle part. Their connecting edges are evaluated with one of seven temporal relations such as "*before*" and "*overlap*," and thus, temporal dependencies in terms of *SS*, *SF*, *FS*, and *FF* values are evaluated, e.g., $SS(atv_8, atv_9) = [1m, 2m]$. Afterward, timed activities are composed to collaborative ones considering their granularities, and modeled in a hierarchical manner as shown in the upper part.

Example 12 (Personalized Detection of Interestingness): Rather than knowing that atv_4 possibly takes a certain time between *[10m, 3h25m]*, analysts may be more convinced of the statement that atv_4 is more likely to be executed within *[2, 3]* hours with a high probability of 90%.

IV. Timed Hierarchical Business Process

In this article, the *B*usiness *P*rocess *M*odel (*BPM*) is mined from an event log, and represented in the *BPMN* format as illustrated in Fig. 2. The widely adopted *I*nductive *M*iner (*IM*) [34], or any other miners, if appropriate, is utilized for this process model mining task. Note that *IM* has been plugged into the open-source framework *ProM*,[1] and implemented upon the open-source process mining platform *PM4Py*.[2] Afterward, temporal constraints relevant to business activities are specified and retrieved from the event log as follows:

1) *Temporal Constraints of Intraactivities* ($TC_{IntraAct}$)*:* Starting time, finishing time and activity durations are obtained from event logs. Specifically, i) starting time is constrained by *SNET*, *SNLT*, or *MSO*, ii) finishing time is constrained by *FNET*, *FNLT*, or *MFO*, and iii) activity durations are formalized and expressed in terms of [*MinT*, *MaxT*] as introduced in Section II.
2) *Temporal Constraints of Interactivities* ($TC_{InterAct}$)*:* Temporal relations are explored through comparing the starting time and finishing time of timed activities. Temporal dependencies are determined in terms of *SS*, *SF*, *FS*, and *FF* values for quantifying timed edges.

Example 13 (Timed Business Process): As illustrated in Fig. 2, timed activities and edges are formalized with temporal constraints. Specifically, timed activities are attached with their starting time, finishing time, and activity durations. Timed edges are specified with temporal relations and dependencies.

Thereafter, extending timed business processes with the notion of temporal granularities is proposed to construct a timed

[1]The inductive miner is available at https://promtools.org
[2]A tutorial of process discovery is available at http://pm4py.org

Algorithm 1: *TH-BPM Discovery.*

Require: - *eLog:* an event log.
- *BPM:* a discovered business process model.

Ensure: - *TH-BPM:* a timed hierarchical *BPM*.

1: $TIA \leftarrow TC_{IntraAct}(eLog,\ BPM)$, $TE \leftarrow TC_{InterAct}(TIA)$
2: **for** each $tia_i \in TIA$ **do**
3: $tia_i.tc.gra \leftarrow \arg\max(GraProb\ (tia_i.tc))$
4: **end for**
5: **for** each $e_i = (tia_i, tia_j) \in TE$ where $tia_i, tia_j \in TIA$ **do**
6: $e_i.tc.gra \leftarrow \arg\max(GraProb\ (e_i.tc))$
7: **end for**
8: $fg \leftarrow \min(T(tia_i).gra)$
9: **while** $\exists\ atv_i\ (\rightarrow / \otimes / \oplus)\ atv_j \in TA \subseteq TIA \cup TCA$ **do**
10: $l_{fg} \leftarrow null$
11: **for** each pair $atv_i\ (\rightarrow / \otimes / \oplus)\ atv_j \in TA$ where $T(atv_i).gra = fg$ or $T(atv_j).gra = fg$ **do**
12: $atv_n \leftarrow atv_i\ (\rightarrow / \otimes / \oplus)\ atv_j$
13: $atv_n \leftarrow TC_{IntraAct}(eLog,\ BPM)$
14: $atv_n.tc.gra \leftarrow \arg\max(GraProb\ (atv_n.tc))$
15: $TCA \leftarrow TCA \cup atv_n$, $TA \leftarrow TA \cup atv_n - \{atv_i, atv_j\}$
16: $l_{fg} \leftarrow l_{fg} \cup atv_i$ if $atv_i.gra = fg$
17: $l_{fg} \leftarrow l_{fg} \cup atv_j$ if $atv_j.gra = fg$
18: $l_{fg} \leftarrow l_{fg} \cup e_i \in TE$ between atv_i and atv_j
19: **end for**
20: $TH\text{-}BPM \leftarrow TH\text{-}BPM \cup l_{fg}$
21: $fg' \leftarrow MappingFuncRel\ (fg, fg') = 1$
22: **end while**

hierarchical business process model. In fact, temporal granularities offer a unique perspective in modeling and analyzing business processes, where analysts can identify which activities (or edges) are more time-consuming (i.e., expressed with coarser granularities) [25]. Besides, temporal granularities provide a distinct approach to integrate business activities. Activities with control relations can be composed into a collaborative one from fine granularities to coarser ones. Therefore, analysts can inspect multiple activities all at once, rather than a single activity multiple times. Note that a collaborative activity commonly takes longer time for its execution and may be expressed with a coarser granularity. Formally, a timed hierarchical business process model can be defined as follows:

Definition 4 (Timed Hierarchical Business Process Model): It is a tuple *TH-BPM = (TIA, TCA, TE, TG)*, where:

1) *TIA* is a set of timed individual activities.
2) *TCA* is a set of timed collaborative activities.
3) *TE* is a set of timed edges between activities where $TE \subseteq (TIA \times TIA) \cup (TIA \times TCA) \cup (TCA \times TCA)$.
4) $TG \subseteq Gra$ is a set of temporal granularities upon the timed activities and edges (i.e., $TIA \cup TCA \cup TE$).

Specifically, a timed hierarchical business process model is constructed by (i) exploring temporal granularities of timed activities (i.e., $TIA \cup TCA$), and timed edges between them (i.e., *TE*), and (ii) composing activities based on their control dependencies from fine granularities to coarser ones. As presented by Algorithm 1, given an event log (denoted *eLog*) and a discovered *BPM*, timed individual activities (i.e., *TIA*) and timed edges (i.e., *TE*) are specified according to afore-presented mining procedures $TC_{IntraAct}$ and $TC_{InterAct}$ (line 1). Each timed individual activity (denoted $tia_i \in TIA$) retrieves its temporal constraints (denoted $tia_i.tc$), such as $T(atv_i)$, and calculates the occurrence probability of each granule (denoted $gra_k \in Gra$) by $GraProb\ (tc) = \sum_{k=1}^{g} (tc.gra = gra_k) \div \sum tc$, where $\sum_{k=1}^{g} (tc.gra = gra_k)$ counts the occurrence times of the granule gra_k, and $\sum tc$ counts that of all granules (lines 2-4). In a similar manner, each timed edge (denoted $e_i \in TE$, where $e_i = (tia_i, tia_j)$ and $tia_i, tia_j \in TIA$) retrieves its most probable temporal granularity of *tc* (lines 5-7). Afterward, the finest granularity (denoted *fg*) is identified (line 8). Activities executed sequently ($\rightarrow$), selectively ($\otimes$) or parallelly ($\oplus$), with the finest granularity *fg* (denoted $atv_i, atv_j \in TA \subseteq TIA \cup TCA$, where $T(atv_i).gra = fg$ or $T(atv_j).gra = fg$), can be composed into a collaborative one (denoted atv_n) (lines 9-12). Thereafter, atv_n is specified with temporal constraints and its granularity (lines 13-14), and is put into *TA* for subsequent composition. Meanwhile, atv_i and atv_j are removed from *TA*, and put into a layer hierarchy (denoted l_{fg}) (lines 15–18).

The procedure specified by lines 11–19 iterates until no activity is available for composition. The layer hierarchy of the proposed model is constructed (line 20), and *fg* is mapped to a coarser one by the function *MappingFuncRel* as introduced in Section II (line 21). This composition process iterates until all activities are composed completely, and a timed hierarchical business process model *TH-BPM* with multiple layers of temporal granularities is constructed. Note that the time complexity of Algorithm 1 is $O(|TA|^2)$, where $|TA|$ is the number of activities in *TH-BPM*. A sample *TH-BPM* is illustrated in the upper part of Fig. 2, as exemplified as follows:

Example 14 (TH-BPM): Temporal granularities are mined for activities in timed business processes as shown in Example 13. For instance, atv_8 and atv_9 are specified with the granularity *minute*, timed edges including *SF*, *SS*, and *FF*(atv_8, atv_9) are specified with the granularity *minute*. Afterward, collaborative activities are formed. atv_8 and atv_9 executed parallelly are composed first, atv_4 and atv_5, atv_6 and atv_7 executed selectively are composed subsequently. Thereafter, atv_2 and ($atv_4 \otimes atv_5$), atv_3 and ($atv_6 \otimes atv_7$) executed sequently are composed. Finally, all individual activities and collaborative ones, are composed, and a *TH-BPM* is generated as shown in Fig. 2. Generally, the majority of activities in *TH-BPM* take minutes for their execution, while atv_4 and atv_6 are more time-consuming in a higher granularity layer (i.e., l_{hour}), whereas atv_3 experiences ignorable latency in seconds.

V. Detection of Temporal Anomalies and Interestingness

This section presents the detection of temporal anomalies and interestingness for timed individual activities and collaborative ones, and their connecting edges upon *TH-BPM*.

A. Temporal Anomaly Detection

Temporal anomalies may occur during the execution of business activities, due to reasons like cyber-attacks or even operation faults. In many scenarios, users may have various acceptance of deviations of anomalies, with respect to different applications, as shown in Example 11. Thus, instead of predefining unchanged signatures that trivially specify the fixed acceptable bounds for anomalous behaviors, this article proposes the time-constrained bounds in a parameterized manner. Personalized acceptance of deviations is enabled by prescribing signatures for detecting temporal anomalies:

Definition 5 (Signature for Temporal Anomaly Detection): It is defined as a tuple $sgt_{ta} =$ (*lb*, *ub*, gra_{pd}, *pn*), where

1) *lb* is an observed lower bound in our *TH-BPM*;
2) *ub* is an observed upper bound in our *TH-BPM*;
3) gra_{pd} is a prescribed granularity for specifying in which granule a deviant execution is acceptable;
4) *pn* is a partitioning number corresponding to gra_{pd}.

Specifically, allowed behaviors for activities and edges are specified by lower and upper bounds *[lb, ub]* observed, i.e., *TIA* ∪ *TCA* ∪ *TE* in *TH-BPM*. For instance, *lb* and *ub* are set to *MinT* and *MaxT* for specifying allowed duration of an activity, while they are set to *SNEL* and *SNLT* for constraining its acceptable starting time, as illustrated in Figs. 1 and 2.

To achieve personalized detection of temporal anomalies, parameterized bounds (denoted *lb (ta)* and *ub (ta)*) are specified from the prescribed signature sgt_{ta}. Specifically, gra_{pd} indicates in which granule a deviant execution is acceptable, and the partitioning number *pn* corresponding to gra_{pd} is specified for refining the bounds. The newly allowed bounds *[lb (ta), ub (ta)]* for temporal anomalies are formulated as:

$$\begin{aligned} \mathbf{max}\ & lb\,(ta) = i \times (G(gra_{pd}) \div pn) \\ \mathbf{min}\ & ub\,(ta) = j \times (G(gra_{pd}) \div pn) \\ \boldsymbol{s.t.}\ & lb\,(ta) \leq lb, \text{ where } i = 0, 1, ..., pn \\ & ub\,(ta) \geq ub, \text{ where } j = 0, 1, ..., pn \end{aligned} \tag{1}$$

where *G*(gra_{pd}) denotes the time duration of the granularity gra_{pd} as introduced in Section II, which can be partitioned into a set of finer intervals, i.e., *seg* = *G*(gra_{pd}) ÷ *pn*. Note that the finer the $sgt_{ta}.gra_{pd}$ is, or the larger the sgt_{ta}.*pn* is, the closer the *[lb (ta), ub (ta)]* to *[lb, ub]*. Thus, temporal anomalies with personalized acceptance of deviations can be effectively detected by parameterizing signatures. For instance, a strict acceptance, such as capturing small variations as anomalies, can be specified by setting a fine gra_{pd} or a large *pn* in sgt_{ta}.

Note that the prescribed granularity gra_{pd} in sgt_{ta} can enhance the efficiency of temporal anomaly detection with a quick preexamination, as elaborated in Algorithm 2. Specifically, a temporal sample (denoted ts) may not be executed between the observed bounds *[lb, ub]*, where a deviation is denoted as *dvt* (lines 1-4). Comparing the granularity of *dvt* (denoted *dvt.gra*) with that of allowed variation (i.e., *Itv*(gra_{pd})*.gra*), we can identify whether such a deviation is acceptable. Specifically, if *dvt.gra* is coarser than *Itv*(gra_{pd})*.gra*, which indicates that *dvt* is largely beyond user's desired range, *ts* is identified as a temporal anomaly (lines 5-6). Otherwise, when *dvt.gra* is finer than *Itv*(gra_{pd})*.gra*, *dvt* is acceptable (lines 7-9). When *dvt.gra* equals to *Itv*(gra_{pd})*.gra*, whether *ts* is an anomaly identified by checking whether it is within the newly allowed bounds *[lb(ta), ub(ta)]* (lines 10-15). Note that executing this algorithm yields liner complexity $O(pn)$, where the time complexity for specifying *[lb(ta), ub(ta)]* in Formula 1 is linear to pn. To summarize, a temporal anomaly can be detected when a temporal sample matches the prescribed signature in following cases.

Algorithm 2: *TAD: Temporal Anomaly Detection.*

Require:-sgt_{ta}: a signature for temporal anomaly detection.
- *ts:* a temporal sample to be detected.

Ensure:- *isAnomaly: True* or *False*.

1: **if** *ts* ∈ *[lb, ub]* **then**
2: **return** *False*
3: **end if**
4: *dvt* ← $\min$ (abs (*ts* - *lb*), abs (*ts* - *ub*))
5: **if** *dvt.gra* > *Itv*(gra_{pd})*.gra* **then**
6: **return** *True*
7: **else if** *dvt.gra* < *Itv*(gra_{pd})*.gra* **then**
8: **return** *False*
9: **end if**
10: *lb (ta), ub (ta)* ← specified by Formula 1.
11: **if** *ts* ∈ *[lb (ta), ub (ta)]* **then**
12: **return** *False*
13: **else**
14: **return** *True*
15: **end if**

Definition 6 (Temporal Anomaly Detection): Given a signature sgt_{ta} and a temporal sample *ts* with respect to timed activities and edges in *TH-BPM*, *ts* is identified as a temporal anomaly when it matches sgt_{ta} in the following cases:

1) *ts* is not within [*lb, ub*], i.e., *ts* < *lb*, or *ts* > *ub*, AND
2) *dvt.gra* > *Itv* (gra_{pd})*.gra*, OR
dvt.gra = *Itv* (gra_{pd})*.gra*, whereas *ts* is not within *[lb (ta), ub (ta)]*, i.e., *ts* < *lb (ta)*, or *ts* > *ub (ta)*.

Otherwise, *ts* cannot be identified as a temporal anomaly when

a) *ts* is within *[lb, ub]*, i.e., *lb* ≤ *ts* ≤ *ub*, OR
b) *dvt.gra* < *Itv* (gra_{pd})*.gra*, OR
c) *ts* lies within *[lb (ta), ub (ta)]*, i.e., *lb (ta)* ≤ *ts* ≤ *ub (ta)*.

Example 15 (Temporal Anomaly Detection): Given a sample signature tuple $sgt_{ta} =$ (*10m*, *3h25m*, *hour*, *24*) corresponding to the activity duration of atv_4 ∈ *TH-BPM.TIA* as shown in Fig. 2, whether atv_4 takes an unacceptable time for its execution is examined, such as *3h40m* and *4h30m*. Specifically, these temporal samples are all out of range *[lb, ub]* = *[10m, 3h25m]* satisfying the case 1 in *Definition* 6, and deviations appear as *dvt* = *15m* and *1h5m* (with the granularity as *minute* and *hour*, respectively). Therefore, (i) *4h30m* is detected as an anomaly, since the granularity of *dvt* is coarser than *Itv(hour).gra* = *minute*, i.e., the signature is matched by the case 2, while (ii) *3h40m* is identified as a nonanomaly satisfying the case (c). More specifically, *dvt.gra* equals *Itv(hour).gra*, and *[lb (ta), ub (ta)]* is refined as elaborated in Formula 1, i.e., *[0,4h]*, where *seg* = *G(hour)* ÷ *24* = *1h*, *i* = *0* and *j* = *4*. We find that *3h40m* is within the newly allowed range *[lb (ta), ub (ta)]*, and thus, cannot be identified as a temporal anomaly. Note that a small variation (e.g., *ts* = *3h40m* where *dvt* = *15m*) can be captured as an anomaly by setting a fine $sgt_{ta}.gra_{pd}$ (e.g., *minute*), or a large $sgt_{ta}.pn$ (e.g., 100).

B. Temporal Interestingness Detection

Temporal interestingness, as the complement to temporal anomalies, represents frequent patterns for activities and edges in *TH-BPM*. As presented in Section III, besides acknowledging that an activity (e.g., atv_4) is executed normally, such as *T*(atv_4) = *3h40m* presented in Example 15, whether it is executed within a refined range (e.g., [2*h*, 3*h*)) with a high probability (e.g., 90%) as illustrated in Example 12, can reveal more information. In this example, *T*(atv_4) is not within the range [2*h*, 3*h*), which means that it appears less frequently, and lies within the range of a probability lower than the prescribed threshold. Specifically, this article parameterizes temporal interestingness by partitioning execution time into a set of intervals with a time-constrained and granularity-aware signature. Formally, a signature for temporal interestingness detection is defined by the following tuple:

Definition 7: (Signature for Temporal Interestingness Detection): It is defined as a tuple sgt_{ti} = (*gra, pn, thrd*), where

1) *gra* is an obtained granularity ∈ *TH-BPM.TG*;
2) *pn* is a partitioning number corresponding to *gra*;
3) *thrd* is a probability threshold for indicating a degree of likelihood of executing within a refined range.

Specifically, the observed execution time (i.e., *[lb, ub]*) of activities and edges in *TH-BPM* can be partitioned into a set of finer time intervals itv_k = *[seg × k, seg × (k + 1))*, where *seg* = *G(gra)* ÷ *pn*, *k* = *0, 1, 2,..., pn* − *1*, *gra* ∈ *TH-BPM* and *pn* refers to a partitioning number corresponding to *gra*. A set of intervals with a total probability larger than the prescribed threshold *thrd* is defined here as temporal interestingness (denoted *lub(ti)*), while formally *lub (ti)* is defined as follows:

$$lub(ti) = \sum_{k=0}^{pn-1} itv_k = \sum_{k=0}^{pn-1} [seg \times k, seg \times (k+1))$$
$$\textbf{\textit{s.t.}} \sum_{k=0}^{pn-1} prob\,p(itv_k) \geq sgt_{ti}.thrd$$
$$\neg\,(seg \times (k+1) \leq lb), \neg\,(ub < seg \times k) \quad (2)$$

where *prob* (itv_k) denotes the probability of executing an activity or an edge within itv_k, and it is calculated as follows:

$$prob\,(itv_k) = \sum itv_k \div \sum_{k=0}^{pn-1} itv_k \quad (3)$$

where $\sum itv_k$ counts the occurrence times of executions within itv_k, and $\sum_{k=0}^{pn-1} itv_k$ counts those in all intervals. For instance, if the duration of an activity atv_4 is within itv_k, i.e., *seg* × *k* ≤ *T*(atv_4) < *seg* × (*k* + 1), the value of $\sum itv_k$ is incremented by one. Note that if itv_k does not overlap with the execution time *[lb, ub]* (i.e., *seg* × (*k* + 1) ≤ *lb*, or *ub* < *seg* × *k*), itv_k can be ignored, since $\sum itv_k = 0$ and *prob* (itv_k) = 0. The calculated probabilities *prob*(itv_k) are ranked and accumulated with larger values until the amount (denoted $prob_{sum} = \sum_{k=0}^{pn-1}$ *prob*(itv_k)) is larger than the prescribed threshold *thrd*. Therefore, the execution time of high possibility are specified for indicating temporal interestingness. Generally, the larger the value of $sgt_{ti}.pn$, the finer the *lub(ti)*. The larger the value of $sgt_{ti}.thrd$ is, the more probable the *lub(ti)* is. Thus, users can parameterize their signatures to achieve personalized performance of temporal interestingness detection.

Algorithm 3: *TID: Temporal Interestingness Detection.*

Require:
- sgt_{ti}: a signature for temporal interestingness detection.
- *ts:* a temporal sample to be detected.

Ensure:- *isInterestingness: True* or *False*.

1: **for** each $itm_i = \{atv_i \in TIA \cup TCA, e_i \in TE\}$ **do**
2: $\quad k \leftarrow itm_i.tc \div seg$
3: $\quad itm_i.tc.prob\,(itv_k)$ += $1 \div \sum itm_i.tc$
4: **end for**
5: **for** each $itm_i \in TIA \cup TCA \cup TE$ **do**
6: $\quad Sort\,(itm_i.tc.prob\,(itv_k))$
7: $\quad$ **while** $itm_i.tc.prob_{sum} < sgt_{ti}.thrd$ and $k < sgt_{ti}.pn$ **do**
8: $\quad\quad itm_i.tc.lub(ti) \leftarrow itm_i.tc.lub\,(ti) \cup itv_k$
9: $\quad\quad itm_i.tc.prob_{sum}$ += $itm_i.tc.prob\,(itv_k)$
10: $\quad\quad k \leftarrow k + 1$
11: $\quad$ **end while**
12: **end for**
13: **if** ∃ $itm_i \in TH\text{-}BPM$ where $ts \in itm_i.tc.lub(ti)$ **then**
14: $\quad$ **return***True*
15: **end if**
16: **return** *False*

Algorithm 3 elaborates the detection procedure for a temporal sample (denoted *ts*) according to a prescribed signature tuple sgt_{ti}. Specifically, ts is evaluated with respect to temporal constraints of timed activities and edges in terms of $itm_i = \{atv_i, e_i\}$ (line 1). The probability of itv_k with respect to a temporal constraint is denoted as $itm_i.tc.prob(itv_k)$, where *k* is obtained through dividing *tc* by *seg* (line 2), and $itm_i.tc.prob(itv_k)$ is accumulated accordingly (line 3). The most probable time intervals (denoted itm_i*.tc.lub(ti)*), i.e., those with a total probability larger than sgt_{ti}*.thrd*, are identified (lines 5–12). Finally, *ts* is checked for whether it is within $itm_i.tc.lub(ti)$ (lines 13–16). Note that this detection is achieved with complexity in size of activities and edges in *TH-BPM*, i.e., $O(|TIA| + |TCA| + |TE|)$. To summarize, temporal interestingness detection can be formulated as follows.

Definition 8 (Temporal Interestingness Detection): Temporal interestingness is detected when a temporal sample *ts* is within *lub(ti)*, which is calculated by Formula 2 according to a prescribed signature sgt_{ti} in Definition 7.

Example 16 (Temporal Interestingness Detection): Given a temporal sample *ts* = *3h40m* with respect to *T*(atv_4) as shown

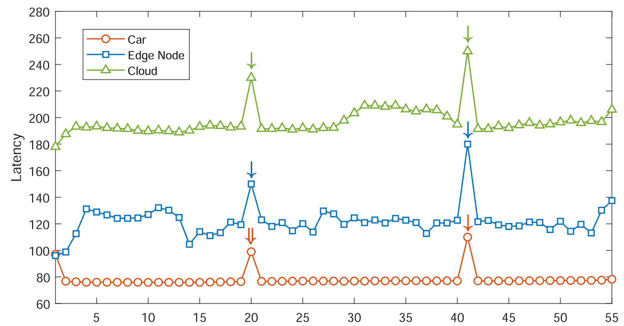


Fig. 3. Latencies of the car, an edge node, and the cloud.

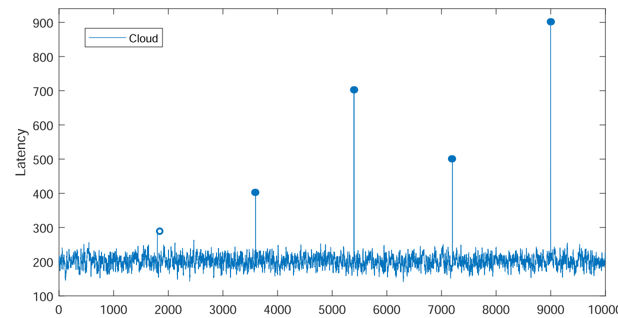


Fig. 4. Latencies of the cloud logged in 10 000 s.

in Example 15, *ts* is identified as a nonanomaly. Furthermore, whether *ts* is executed in intervals of a high probability like 80% by a signature sgt_{ti} = (*hour*, 24, 80%) can also be identified. Specifically, a duration with the granularity *hour* (i.e., *G(hour)* = 24 h) is partitioned into 24 segments (i.e., *seg* = 1 h). The activity duration (i.e., *[lb, ub]* = [10*m*, 3*h*25*m*]) is partitioned into a set of finer intervals, i.e., itv_0 = [0, 1*h*), itv_1 = [1*h*, 2*h*), itv_2 = [2*h*, 3*h*), and itv_3 = [3*h*, 4*h*). *lub(ti)* is calculated by Formula 2 (e.g., *lub(ti)* = $\{itv_1, itv_2\}$ = [1*h*, 3*h*)), and *ts* $\notin$ *lub(ti)* is not identified as temporal interestingness.

Example 17 (Temporal Interestingness Detection): For another temporal sample *ts* = *1h30m*, *ts* is identified as normal to satisfy the case (a) in Definition 6. Specifically, *ts* $\in$ *lub(ti)* is identified as temporal interestingness. When parameterizing the threshold sgt_{ti}*.thrd* to a higher value, e.g., 90%, *lub(ti)* is specified as a finer interval, e.g., [2*h*, 3*h*), and *ts* $\notin$ *lub(ti)* is not identified as temporal interestingness.

VI. Implementation and Evaluation

A prototype has been implemented in Python, and experiments have been conducted upon a desktop with an Intel(R) Core(TM) i5-28250U, CPU at 1.60 GHz and 1.80 GHz, a 8-GB memory and a 64-bit operating system.

A. Case Study

To evaluate the applicability of our technique, a case study has been conducted with *IoT* devices deployed in an edge network,[3] and latencies are collected real-timely:

1) *Network Deployment:* An edge network is simulated with various *IoT* devices including automobiles, edge nodes, and the remote Cloud [30]. Self-driving cars ride on the roads in a smart city and route their data to edge nodes, while edge nodes communicate with the Cloud.
2) *Event Logs:* Latencies of a car, an edge node, and the Cloud are recorded for a total of 10 000 seconds (over 2.5 h), as shown in Figs. 3 and 4, where the latencies are of the granularity *millisecond*.

Generally, these figures show that these devices have different latencies. Specifically, the car has the lowest latencies ranging from 50 to 100 ms, i.e., [*lb, ub*] = *[50 ms, 100 ms]*, while the edge node and the Cloud experience higher latencies, i.e., *[lb, ub]* = *[100 ms, 300 ms]*.

Temporal anomalies and interestingness are detected from the logged latencies, where temporal variations are simulated by randomly setting much higher latencies. These latencies (marked by ↓ and ⇓ in Fig. 3, ○ and ● in Fig. 4) can be identified as temporal anomalies by parameterizing gra_{pd} = *millisecond* and *pn* = 50, such as sgt_{ta} = (*50 ms*, *100 ms*, *millisecond*, 50). Giving sgt_{ti} = (*millisecond*, 50, 80%), *lub(ti)* is specified, such as *[180 ms, 220 ms]* for the Cloud, latencies within the range are identified as temporal interestingness. To summarize, the detection of temporal anomalies and interestingness can be achieved in this edge scenario, and be used to identify rare events and probable behaviors in a complementary manner.

In fact, the detection of temporal anomalies and interestingness is parameterized by setting different signatures. For instance, setting sgt_{ta} = (*50 ms*, *100 ms*, *second*, 50) can identify the latency of the car at the *20th* point (denoted ts_{car}^{20}) as normal. Note that a larger tolerance of anomalous behaviors can be reflected by a coarser gra_{pd}, or a smaller *pn* in sgt_{ta}. The latency of the Cloud at the *2000th* point can be alternatively identified as normal, by setting a coarser granularity sgt_{ta}*.gra* = *second*. By parameterizing a smaller *pn* in sgt_{ta}, such as sgt_{ta} = (*50 ms*, *100 ms*, *millisecond*, 20), ts_{car}^{20} can be identified as a normal latency as well.

B. Performance Metrics

The performance of our technique is evaluated from aspects of *accuracy*, *efficiency,* and *generality*. Specifically, three kinds of commonly adopted performance metrics, i.e., *accuracy*, *precision*, and *recall*, are calculated for accuracy evaluation:

$$acc_{ta} = \frac{TP_{ta} + TN_{nta}}{P_{ta} + N_{nta}},\ acc_{ti} = \frac{TP_{ti} + TN_{nti}}{P_{ti} + N_{nti}} \tag{4}$$

$$pre_{ta} = \frac{TP_{ta}}{TP_{ta} + FP_{ta}},\ pre_{ti} = \frac{TP_{ti}}{TP_{ti} + FP_{ti}} \tag{5}$$

$$rec_{ta} = \frac{TP_{ta}}{TP_{ta} + FN_{nta}},\ rec_{ti} = \frac{TP_{ti}}{TP_{ti} + FN_{nti}} \tag{6}$$

where acc_{ta}, pre_{ta} and rec_{ta} denote the accuracy, precision and recall for temporal anomaly detection, while acc_{ti}, pre_{ti}, and rec_{ti} denote that for temporal interestingness detection. P_{ta} refers to the number of anomalies, which include those identified correctly (denoted TP_{ta}, e.g., the anomalous duration *4h30m* in

[3]A demo video is available at https://vimeo.com/manage/videos/595793479

TABLE I
DESCRIPTIVE STATISTICS OF PUBLICLY-AVAILABLE REAL-LIFE EVENT LOGS IN TERMS OF TRACES AND EVENTS

Abbreviated Log Name	Total Traces	Total Events	Event Classes	Trace Length		
				Minimum	Mean	Maximum
uci_workdays	25	1392	32	34	56	92
uci_weekends	10	488	30	28	49	70
102_workdays	18	1152	36	46	64	82
102_weekends	7	420	36	46	60	76
104_workdays	43	4200	38	58	98	134
104_weekends	18	1728	38	56	96	124
110_workdays	21	1390	34	20	66	90
110_weekends	6	368	28	54	61	66

Example 15) and those failed to be detected (denoted FN_{nta}, e.g., if *4h30m* in Example 15 is identified as a nonanomaly), i.e., $P_{ta} = TP_{ta} + FN_{nta}$. In a similar fashion, N_{nta} refers to the number of nonanomalies, which may be identified correctly (denoted TN_{nta}, e.g., the nonanomalous execution time *3h25m* in Example 15), or identified incorrectly as anomalies (denoted FP_{ta}, e.g., if *3h25m* in Example 15 is identified as an anomaly), i.e., $N_{ta} = TN_{nta} + FP_{ta}$. Besides, the number of interestingness is denoted by $P_{ti} = TP_{ti} + FN_{nti}$, where TP_{ti} stands for the number of cases when interestingness is identified correctly, and FN_{nti} counts the cases when interestingness is identified incorrectly. N_{ti} denotes the number of noninterestingness, which may be identified correctly (denoted TN_{nti}), or identified incorrectly as interestingness (denoted FP_{ti}), i.e., $N_{ti} = TN_{nti} + FP_{ti}$.

Efficiency is another key indicator for performance evaluation, especially in *IoT*-based edge environments. Anomalies and interestingness are expected to be detected as soon as possible to infer cyber-attacks in time, even when the calculation of the model is high, whereas resources of edge nodes are mostly scarce. In our experiments, the execution time for detecting temporal anomalies and interestingness is recorded.

Besides, the *generality* of our *TH-BPM* is emphasized in different scenarios. Theoretically, our work can be applied upon any event logs with time stamps recorded for business activities, without additional limitations or assumptions. In our experiments, extensive evaluations upon heterogeneous datasets, rather than a single or homogeneous one(s), are conducted to demonstrate the generality of our technique. Specifically, i) latencies of different *IoT* devices recorded in a real edge network as presented in Section VI-A, and ii) real-life event logs publicly available at *4TU Centre* as presented in Section VI-C, are adopted for the performance evaluation.

C. Event Logs at 4TU Center

The *4TU Center for Research Data*[4] is used for evaluating our technique. *Activities of daily living of several individuals logs*[5] are logged by *Fluxicon Disco*,[6] such as *sleeping*, *meal preparation*, and *washing* recorded on *workdays* and at *weekends*. The event logs show not only different behaviors of individuals but also common patterns. Table I shows the details of these event logs, such as the log size, ranging from 6 (*110_weekends*) to 43 traces (*104_workdays*), and the number of events, varying from 368 to 4200. Temporal constraints and granularities are retrieved from event logs. Specifically, the event logs consist of trace names, event names, time stamps, and transitions (i.e., *start* and *complete*). Other event logs with time information recorded can be applied as well. Anomalies exist in these event logs due to the casualness of activities, while interestingness can be identified from the routines of activities. For instance, individuals commonly *sleep* at a regular time period but in a casual manner.

D. Baseline Techniques

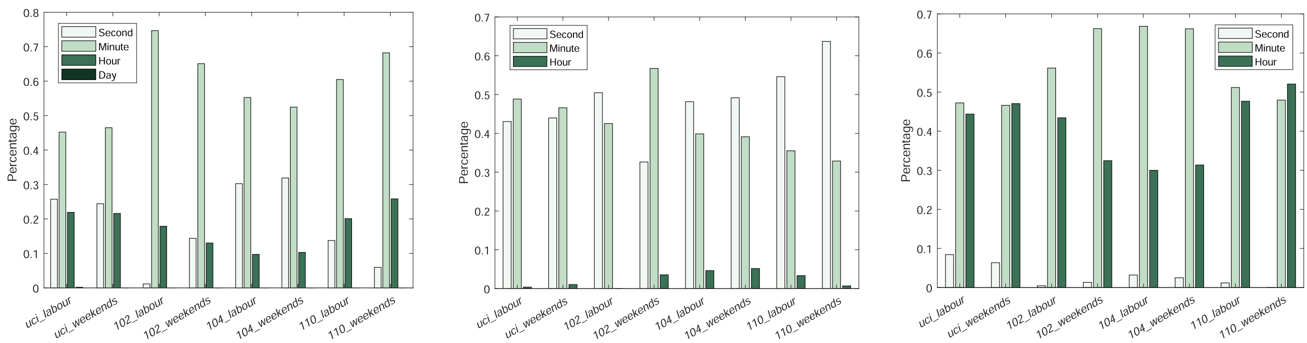
The following approaches are selected as baseline techniques [35] for the comparison with our *TH-BPM:*

1) *K*-*N*earest *N*eighbor (*KNN*) detects anomalies with the *k*th nearest neighbor by evaluating outlying scores [36].
2) *L*ocal *O*utlier *F*actor (*LOF*) measures anomaly scores according to the isolation degree of a certain sample with respect to its neighbors [37].
3) *M*inimum *C*ovariance *D*eterminant (*MCD*) estimates anomalies by computing the Mahalanobis distance as the outlier degree of datasets [38].
4) *H*istogram-*B*ased *O*utlier *S*core (*HBOS*) extracts the feature independence and calculates the degree of outlyingness by building histograms [39].

E. Experimental Settings

In our experiments, we shuffle event logs and adopt 80% for model discovery, while the remaining 20% for evaluation purpose. Signatures for anomaly and interestingness detection are set as follows: i) sgt_{ta}: $sgt_{ta}.gra_{pd}$ is set to a value finer than its obtained granularity in *TH-BPM.TG*, and partitioning numbers $sgt_{ta}.pn$ for granularity *second*, *minute*, *hour*, and *day* are set to 6, 6, 24, and 7, respectively, as is illustrated by Examples 15–17, and (ii) sgt_{ti}: $sgt_{ti}.pn$ is set to the same value as that of $sgt_{ta}.pn$, and $sgt_{ti}.thrd$ is set to 70%. Note that other values can also be used if appropriate. The probability threshold of outlier degrees is prescribed as 5% for baseline techniques. Temporal interestingness detection is the complement to anomaly detection, such that a relative high threshold (e.g., 90%) can be interpreted as the degree of likelihood to identify frequent behaviors. Note that the ground truth of anomalies and interestingness (i.e., P_{ta} and P_{ti}

[4]*4TU.ResearchData* is available at http://data.4tu.nl
[5]https://doi.org/10.4121/uuid:01eaba9f-d3ed-4e04-9945-b8b302764176
[6]*Fluxicon Disco* is available at http://fluxicon.com/disco



(a) (b) (c)

Fig. 5. Average percentages of temporal granularities of individual activities' durations, edges between activities and execution time of collaborative activities, where (i) the majority of individual activities are executed in minutes, (ii) most of edges between activities take seconds to minutes, while (iii) collaborative ones are mostly expressed with granularities of *minute* and *hour*.(a) Individual activities. durations. (b) Edges between activities. (c) Collaborative activities. execution time.

as introduced in Section VI-B) is assumed as what is identified by the widely adopted *KNN* method and meanwhile matches the prescribed signatures satisfying users' personalized demands.

F. Evaluation Results

Experiments are conducted upon aforementioned datasets, and evaluation results in comparison with baseline techniques are presented in the following sections.

1) Comparison Upon Temporal Granularities: Temporal granularities of individual activities, collaborative ones and their connecting edges are explored upon the event logs depicted in Table I. The average percentages of these granularities in terms of durations of individual activities are shown in Fig. 5(a), intervals of their connecting edges are illustrated in Fig. 5(b), and total execution time of collaborative activities are presented in Fig. 5(c). The results show that the majority of individual activities are executed in minutes, while collaborative ones are more likely to take minutes to hours. Edges between activities are mostly expressed with granularities of *second* and *minute*. Besides, the latencies logged in our deployed edge network show that the *IoT* devices generally take milliseconds for their executions. These results demonstrate that temporal granularities vary in business activities and edges in *TH-BPM*.

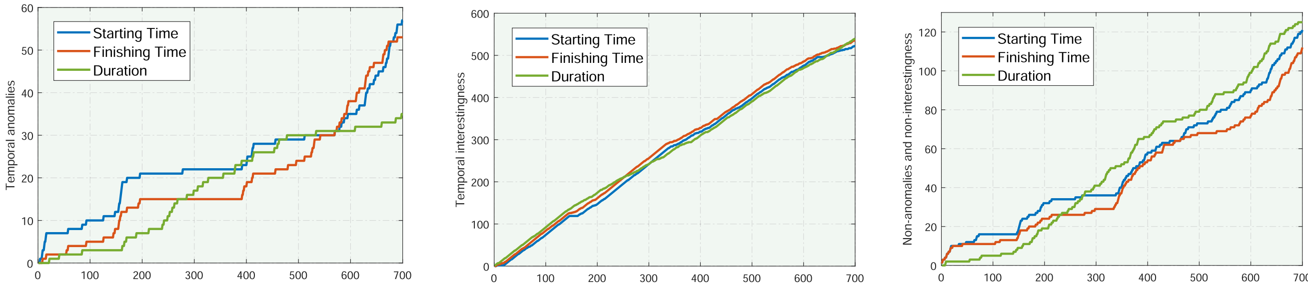
Generally, the obtained granularities can reveal meaningful temporal executions to analysts from a general perspective. For instance, the execution time of the majority of individual activities and those of collaborative ones as shown in Fig. 5(a) and (c), express different temporal granularities. Intuitively, collaborative activities take longer time than that of individual ones, and commonly express coarser granularities. These temporal granularities in *TH-BPM* are modeled as sampled in Fig. 2, which facilitate the inspection of business activities, especially those collaborative ones that are largely unexplored in the current literature. To summarize, the results demonstrate the applicability and effectiveness of temporal granularities, especially when business processes vary significantly in their granules. This means that granularity-aware signatures can improve the efficiency of temporal anomaly and interestingness detection.

2) Comparison Upon Temporal Constraints: As shown by Fig. 5, relative and absolute temporal constraints are explored with respect to individual activities and collaborative ones, and their connecting edges in *TH-BPM*. The results show that different temporal constraints can be acquired from event logs and expressed with certain temporal granularities. In this experiment, different temporal constraints including starting time, finishing time, and durations of activities as introduced in Section II, are studied as shown in Figs. 6, 7, and 8.

Specifically, the datasets are emulated as they arrive in real time to evaluate runtime detection, and demonstrate the applicability of our *TH-BPM* in real scenarios. Fig. 6 presents the numbers of activities that are identified as anomalies, interestingness, etc., respectively. The results shown in Fig. 6(a) and (b) indicate that the acquired temporal constraints, i.e., starting time, finishing time, and activity durations are more likely to appear as temporal interestingness rather than temporal anomalies. The timed activities are less likely executed with unacceptable deviations (i.e., as anomalies), but more possibly executed in certain intervals of a high possibility (i.e., as temporal interestingness). Besides, nonanomalies and noninterestingness exist in our scenarios as shown in Fig. 6(c). To summarize, the results demonstrate that our technique is applicable in real-time detection scenarios.

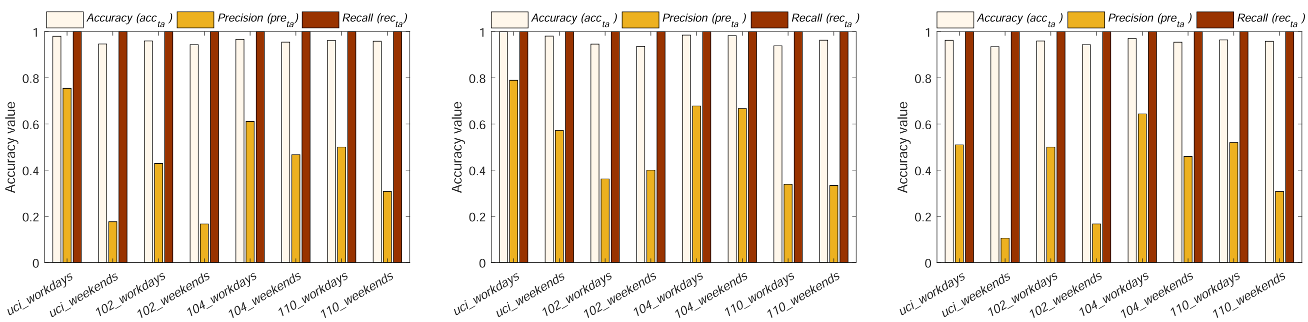
Temporal anomalies and interestingness are evaluated using performance metrics, including *accuracy*, *precision*, and *recall* as presented in Formulas 4, 5, and 6. The performance metrics vary with different event logs, and demonstrate that this technique can be applied in heterogeneous scenarios with high accuracy values. Specifically, the accuracy metrics of temporal interestingness are mostly larger than that of temporal anomalies. This result conforms with what is shown in Fig. 6, which indicates that temporal anomalies and interestingness are detected in a complementary manner. The more anomalous behaviors are detected, the less likely the business activities are executed in their probable time intervals, and *vice versa*.

As a result, the technique is applicable and effective in detecting anomalous behaviors and probable executions from the time perspective. In fact, a more comprehensive temporal modeling is provided in this article to facilitate the detection of



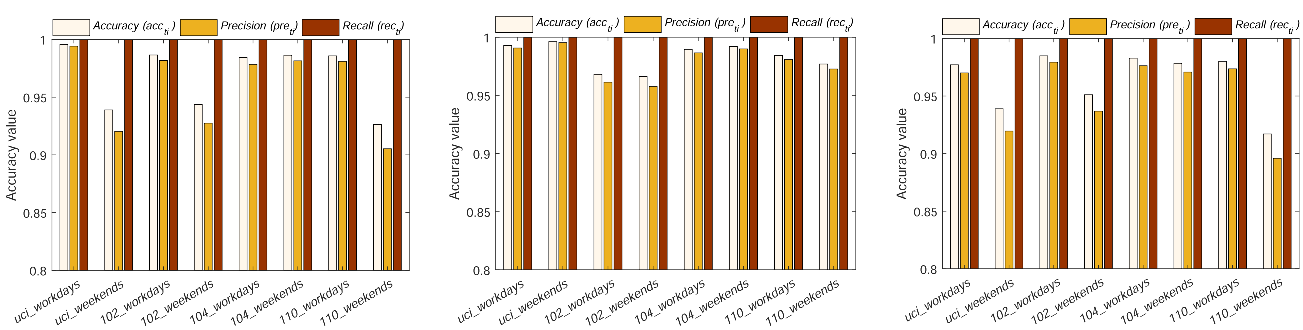
(a) (b) (c)

Fig. 6. Runtime detection of temporal anomalies and interestingness with an event log (i.e., *uci _ workdays*) emulated in real time, where (i) more anomalies are detected for starting and finishing time than that for activity duration, (ii) these temporal constraints are more likely to appear as temporal interestingness than anomalies, and (iii) nonanomalies and noninterestingness exist in our scenarios.(a) Temporal anomalies. (b) Temporal interestingness. (c) Non-anomalies and non-interestingness.



(a) (b) (c)

Fig. 7. Average *accuracy* (acc_{ta}), *precision* (pre_{ta}) and *recall* (rec_{ta}) of temporal anomaly detection in terms of starting time, finishing time and activity durations, where *accuracy* and *recall* are mostly larger than 90%, while *precision* varies from 10% to 88% against different event logs.(a) Starting time. (b) Finishing time. (c) Activity durations.



(a) (b) (c)

Fig. 8. Average *accuracy* (acc_{ti}), *precision* (pre_{ti}) and *recall* (rec_{ti}) of temporal interestingness in terms of starting time, finishing time, and activity durations, which are mostly larger than 90% and than that of temporal anomalies. This result shows that these temporal constraints are more likely to display as interestingness than anomalies, which indicates the complementarity between temporal anomalies and interestingness.(a) Starting time. (b) Finishing time. (c) Activity durations.

temporal anomalies and interestingness. Leveraging prescribed signatures, this technique achieves better performance in terms of various temporal constraints with higher accuracy, as shown by the results of these experiments.

3) Accuracy Comparison: To evaluate from the aspect of accuracy, this section presents the average values of *accuracy*, *precision*, and *recall* for temporal anomaly and interestingness detection. Fig. 9 shows the average values of relevant performance metrics for temporal anomaly detection with various event logs. The performance metrics vary with techniques utilized and event logs. Generally, the *accuracy* (i.e., acc_{ta}) shows better performance (above 85% mostly), while the *precision* and *recall* of baseline techniques decrease due to dissatisfaction of user's customized demands. In contrast, *TH-BPM* tackles this issue by parameterizing signatures to satisfy personalized demands, and achieves the best performance in most scenarios with an improvement of accuracy by 6.08%. Specifically, *KNN* and *MCD* outperform other solutions with

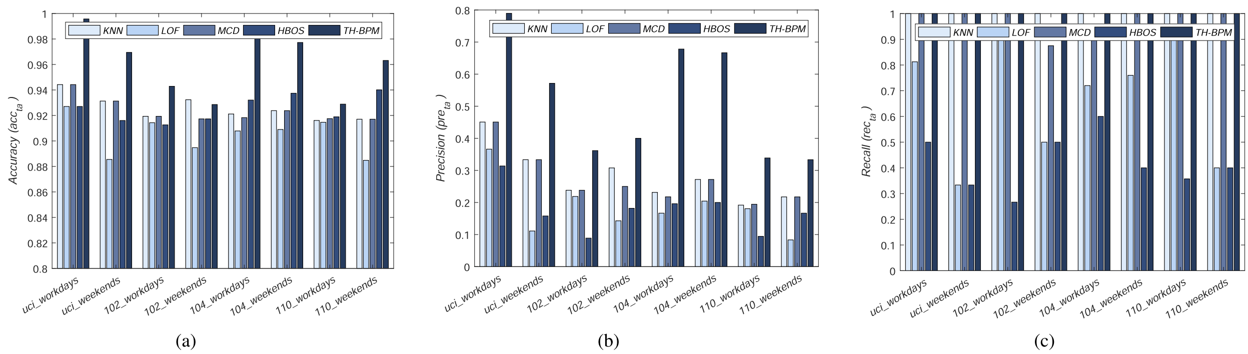


Fig. 9. Average *accuracy* (acc_{ta}), *precision* (pre_{ta}), and *recall* (rec_{ta}) of temporal anomaly detection upon various event logs, where (i) values of accuracy are mostly above 85% for these techniques, (ii) *KNN*, *MCD*, and our *TH-BPM* achieve better performance in terms of *precision* and *recall*, and (iii) our *TH-BPM* outperforms these baseline techniques in terms of average *accuracy*, *precision*, and *recall*, and improves accuracy by 6.08%.(a) Accuracy (accta). (b) Precision (preta). (c) Recall (recta).

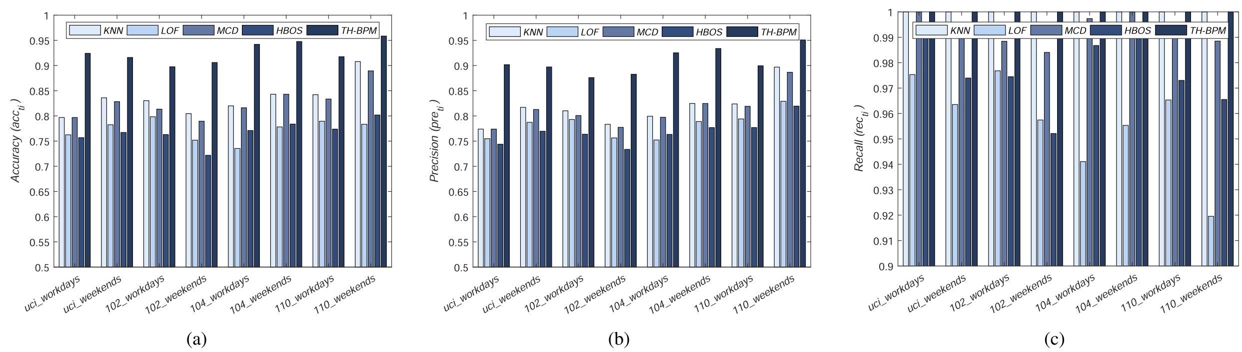


Fig. 10. Average *accuracy* (acc_{ti}), *precision* (pre_{ti}), and *recall* (rec_{ti}) of temporal interestingness with different algorithms on various event logs, where (i) average *accuracy* and *precision* are all above 65%, and *recall* above 85%, (ii) *KNN*, *MCD*, and *TH-BPM* achieve better performance, and (iii) our *TH-BPM* outperforms these baseline techniques with its performance metrics.(a) Accuracy (accti). (b) Precision (preti). (c) Recall (recti).

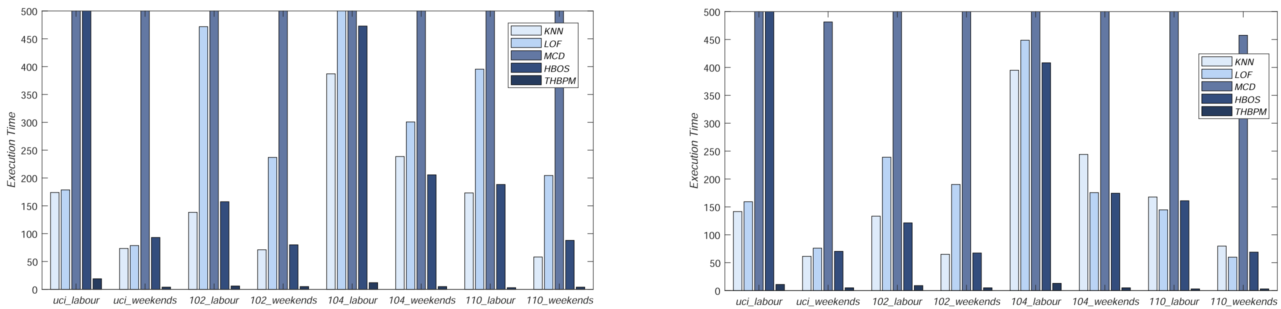
larger values of relevant performance metrics. Generally, comparison results show different outperformance upon heterogeneous event logs, which demonstrate the applicability and generality of this technique in pervasive scenarios.

The average *accuracy*, *precision*, and *recall* for temporal interestingness detection are illustrated in Fig. 10. This figure shows that our technique achieves better performance than baseline techniques with higher accuracy. Besides, the performance metrics for temporal interestingness detection express larger values than those for anomaly detection. This fact indicates that more interestingness is identified rather than anomalies upon these event logs, and it is consistent with other evaluation results as shown in Figs. 6–8. Generally, the baseline techniques analyze activity behaviors from historical data, by measuring outlying scores according to statistical correlation and data distribution, without consideration of parameterized signatures for personalized satisfaction, while *TH-BPM* is more applicable in detecting temporal anomalies and interestingness with higher accuracy, and wider application in heterogenous event logs.

4) Efficiency Comparison: To evaluate and compare the efficiency of our technique with baseline techniques, total execution time for detecting temporal anomalies and interestingness with various event logs is recorded as shown in Fig. 11. Generally, *KNN* takes shorter time among baseline techniques in most scenarios, whereas *MCD* underperforms with over 500 ms for its detection mostly. Our *TH-BPM* is less time-consuming (shorter than 50 ms), and more efficient in detecting temporal anomalies and interestingness in these scenarios. Besides, the efficiency of our algorithms is also evaluated by a real edge network for detecting anomalous and probable latencies, as elaborated in Section VI-A. To summarize, by proposing *TH-BPM* and pattern-based techniques with parameterized signatures, our technique performs much quicker than other solutions in various parameter settings.

5) Comparison of Temporal Modeling: As discussed in Section VII, efforts have been proposed to provide formalized temporal constraints for business processes, which are ensured to adhere to these constraints. Table II presents a comparison of current research approaches against our *TH-BPM*, regarding supported temporal constraints and granularities that are categorized as follows:

1) *Intraactivity:* starting time, finishing time, and duration of an individual activity.
2) *Interactivity:* interval, temporal relation, and temporal dependence between activities.
3) *Collaborative activity:* starting time, finishing time, and the total execution time of a collaborative activity.



(a) (b)

Fig. 11. Total execution time of temporal anomalies and interestingness detection with different techniques on various event logs, where (i) *MCD* generally takes more than 500 milliseconds, (ii) *KNN* is less time-consuming in comparison with other baseline techniques, while (iii) our *TH-BPM* outperforms baseline techniques with the smallest execution time (less than 50 milliseconds).(a) Execution time of temporal anomaly detection. (b) Execution time of temporal interestingness detection.

TABLE II
A Comparative Result Regarding to Temporal Constraints and Granularities in Related Techniques as Discussed in Section VII

Approaches [Ref.]	Intraactivity		Interactivity			Collaborative Activity	Temporal Granularity
	Duration	Starting / Finishing Time	Interval	Temporal Relation	Temporal Dependence		
Wang *et al.* [9]	✓	✓	✓	✓	✓	✓	✓
Zhang *et al.* [13]	✓	✓	✓	✓	✓	✓	×
Halima *et al.* [16]	✓	✓	✓	✓	✓	×	×
Sun *et al.* [17]	✓	✓	✓	×	×	×	×
Senderovich *et al.* [18]	✓	×	✓	✓	×	×	×
Liu *et al.* [19]	✓	✓	×	×	×	×	×
Rachidi *et al.* [20]	✓	×	✓	×	✓	×	×
Pereira *et al.* [21]	✓	✓	✓	×	✓	✓	×
Morales *et al.* [22]	✓	×	×	×	×	×	×
Kumar *et al.* [23]	✓	×	×	×	✓	×	×
Lanz *et al.* [40]	✓	✓	✓	×	✓	×	×
Our *TH-BPM*	✓	✓	✓	✓	✓	✓	✓

4) *Temporal granularity:* calendar-based granularities.

Table II suggests that most techniques attempt to explore partial temporal constraints to achieve certain temporal analysis objectives. When it is based on limited temporal constraints, temporal analysis, such as time conflict analysis, would also be limited. Besides, few studies take temporal granularity into consideration. Therefore, the conclusion can be made that *TH-BPM* is capable of providing a more comprehensive temporal modeling, and the detection of temporal anomalies and interestingness upon the acquired timed activities and edges is supposed to be more sound and effective.

Specifically, the work proposed in [9] achieves a solid temporal modeling, but it falls short in satisfying personalized demands for the detection of temporal anomalies and interestingness. To fill this gap, this article proposes signature-based detection for temporal anomalies and interestingness, where signatures can be prescribed and parameterized to achieve a high level of flexibility and applicability.

To summarize, this technique achieves better performance than baselines in various parameter settings, when factors including *accuracy*, *efficiency*, and *generality* are considered. Specifically, personalized demands are emphasized in this setting, and satisfied by parameterizing signatures for temporal anomalies and interestingness. Our *TH-BPM* improves accuracy by 6.08%, and reduces execution time to less than 50 ms, with evaluations in heterogenous scenarios.

VII. Related Work

This section provides an overview of research issues for anomaly and interestingness detection, and related works for temporal modeling of constraints and granularities.

A. Anomaly Detection

Anomaly detection is a promising research problem that has been studied in many research and application areas [35]. Researchers continue to provide solutions for efficient anomaly detection. Various discovery algorithms and measurement design have been proposed in recent years [2], [6]. Generally, reference models for describing normal behaviors are mined from event logs, and anomalous behaviors deviating from the model can be identified. For instance, Myers *et al.* [41] proposed an approach to identify anomalous behaviors, and design a conformance checking analysis technique to determine the extent to which real behaviors obtained in device logs match expected behaviors captured in a process model. A similar anomaly detection approach is developed in [42] through combining three different types of approaches with a creative computing method to increase efficiency and improve accuracy. To automatically detect and analyze anomalies, an autoencoder is introduced in [31]. Generally, studies aim to mine normal behaviors from event logs, and detect behaviors that deviate from normal ones as anomalies. However, less considerations were given to personalized demands of the users. Taking this issue into account, this article allows the parameterization of signatures to better satisfy user's demands in real scenarios.

Besides detecting anomalies using traditional techniques, learning-based algorithms have been developed through supervised and unsupervised methods [29], [43]. Various unsupervised algorithms have been developed [44], as mentioned in Section VI-D, to discover anomalies by measuring outlying scores and identifying significant deviations as anomalies. However, these algorithms suffer from model instability due to their unsupervised nature, and constantly face higher false positive rate. Supervised methods leveraging deep learning have also been investigated, such as a neural network architecture presented in [29] for real-time process-aware anomaly monitoring. However, due to the complexity of deep models, they usually require a large amount of computing resources, and can hardly be applied to edge *IoT* devices, which are less intelligent and generally have limited capacities. Instead, the signature-based strategy shows significant advantages on a more effective detection, and is applicable in edge scenarios demonstrated in Section VI-A.

To summarize, current efforts mainly explore anomalies leveraging outlyingness- and signature-based methods from perspectives, such as control- and data-flow perspectives [45]. However, anomaly detection from the time dimension has not been explored extensively. Specifically, anomalous behaviors with respect to individual business activities and their connecting edges are studied, whereas collaborative activities have not been well explored. Besides, personalized demands have rarely been investigated, such as different acceptance of deviations for anomaly detection [9]. Consequently, this article attempts to tackle these issues by parameterizing the anomaly detection with respect to extensive temporal constraints, and putting forward time-constrained and granularity-aware signatures to specify personalized acceptance of deviant executions for anomalous behaviors.

B. Interestingness Detection

Interestingness detection, as the complement to anomaly detection, has been proposed to identify various types of interestingness with respect to user's knowledge and concern [9], [46]. For instance, interesting subsets of cases that have deviations or that are delayed in process mining are modeled in [8]. A systematic review and comparative evaluation of sequence classification techniques for business process deviance mining is presented in [47], leveraging frequent and discriminative pattern mining, where the interestingness measure serves as a proxy for the usefulness of extracted patterns. To automatically identify interesting process behaviors related to control- and data-flow perspectives, an approach named *ProcessExplorer* is proposed in [48] for generating appropriate subset recommendations. In [49], authors introduce the specification and verification of reactive constraints, and devise a time evaluation method to facilitate the recognition of interestingness of discovered constraints. Generally, these methods can effectively evaluate interestingness from different aspects [11]. Following this trend, this article proposes to explore interesting patterns from the time dimension, i.e., temporal interestingness, which is largely unexplored. The most frequent patterns are identified leveraging time-constrained and granularity-aware signatures, where the execution time of business activities is parameterized in user-defined granules and ranked by their probabilities.

Besides, learning-based methods have been recently applied, such as unsupervised interestingness detection algorithms [35], like *KNN*, *LOF*, *MCD*, and *HBOS*, as presented in Section VI-D. Generally, these methods compute the degree of interestingness from outlying scores of anomalies. The smaller the outlying scores, the larger the degree of interestingness. However, they require a training phase to develop the database of general behaviors, and may hardly be applicable in edge scenarios. Besides, a careful setting of threshold level of detection makes them complex. Specifically, deep neural networks have also been designed for detecting interestingness tasks, such as predicting image and video visual interestingness [46] and media interestingness [50]. However, these methods require relatively large capacities and resources to work properly, which may induce unaffordable burden and undesired latency for edge *IoT* devices. Besides, few efforts have been made in study of personalized demands mentioned in related works of anomaly detection. Generally, these learning-based methods have pros and cons from different aspects.

To summarize, current techniques for interestingness detection mainly concern with patterns, classifiers, and subsets of cases [3], while temporal information and parameterized measurement are seldom considered during the interestingness assessment. To fill this gap, our approach evaluates temporal interestingness as the complement to temporal anomalies, where signatures are designed for modeling and recommending interestingness in a parameterized manner. Besides, our proposed approach is more applicable and effective than those learning-based methods, especially in edge scenarios.

C. Temporal Modeling

Techniques have been developed to support the exploitation and utilization of various temporal constraints from event logs. For instance, the allocation of time-aware Cloud resource in business processes is proposed in [16], which investigates relative and absolute temporal constraints. Temporal constraints, including starting and finishing time, and activity duration, are considered for service composition in timed *IoT* applications [17]. A temporal network representation of event logs is introduced in [18], which mainly captures delays and temporal relations between activities. Another work concerning temporal network representation is proposed in [19] via time-reinforced random walk, where temporal nodes, edges, and networks are defined, to represent a time-evolving graph with duration information in real systems. Temporal constraints of interactivities are concerned in [20] for a petrinets-based modeling of discrete event systems. Besides, temporal constraints in the management of business processes including durations, intervals, and temporal dependencies, have also been studied [21]. Generally, these approaches aim to address various challenging tasks by utilizing and transferring temporal constraints and their derivatives, but they fail to explore temporal constraints comprehensively. Comparisons of current techniques with our *TH-BPM* are presented in Table II, and they demonstrate that our *TH-BPM* outperforms these techniques in terms of temporal modeling.

Generally, analyzing time-constrained business processes becomes increasingly important in many real applications [21]. For instance, *BPMN* models generally lack formal semantics to conduct qualitative analysis, while transforming them to timed automata networks allows analysts to perform the evaluation with respect to business performance indicators (e.g., service time) driven from business needs [22]. Managing violations of temporal process constraints is of significance as introduced in [23], where a schedule optimization approach is developed. Besides, introducing semantic for time patterns enables the integration of existing process-aware information systems [40]. Taking these into consideration, this article proposes to evaluate temporal anomalies and interestingness based on an extensive modeling of temporal constraints and granularities.

VIII. Conclusion

This article constructs a timed business process model with hierarchical granularities for temporal modeling and analysis, and conducts the detection of anomalies and interestingness from the time perspective. Generally, (i) temporal constraints, including starting and finishing time of intraactivities, activity duration, temporal relations, and dependencies of interactivities, and (ii) temporal granularities, with respect to individual activities and collaborative ones, and their connecting edges, are formalized. A timed hierarchical business process model is proposed by composing activities from fine granularities to coarser ones. Thereafter, temporal anomalies and interestingness are detected, where signature-based deviation tolerance and probable interval magnitude are specified according to users' personalized requirements. Extensive experiments are conducted upon publicly available real-life event logs. Evaluation results show the outperformance of our approach in terms of accuracy, efficiency, and generality, in comparison with the state-of-the-art techniques.

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Managerial Relevance Statement: To promote the protection of cyber-threats, this article probes potential cyber-attacks, where anomaly detection, serving as an efficient means of probing cyber-attacks upon business processes, is proposed for the identification of unexpected behaviors in the recent literature. Besides, interestingness detection, which is complementary to anomaly detection, is conducted to manifest the most frequent behaviors of business processes. Conventional solutions for the detection of anomalies and interestingness are insufficient, especially from a time dimension. They further fall short of satisfying personalized demands of end users. To solve these issues, this article constructs a timed business process model with hierarchical granularities for temporal modeling and analysis, and conducts a customized detection for temporal anomalies and interestingness leveraging time-constrained and granularity-aware signatures. The effectiveness of the proposed approach is evidenced through a case study, and further extensive experiments, on public datasets. Evaluation results show that our approach performs better than baseline techniques in terms of accuracy, efficiency, and generality, and demonstrate the applicability in real-world scenarios.

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